

Non-viral transfection of animal tissue culture cells

Non-viral transfection introduces DNA or RNA into animal tissue culture cells using chemical delivery methods. The approach avoids viral vectors and enables rapid, scalable manipulation of a wide range of cell types to conduct transient expressions, reporter assays, and genetic perturbation screens.

Lipofectamine® reagents (*Alternative A*) are mixtures of cationic lipids (DOSPA) and neutral helper lipids (DOPE) that form liposomes, which fuse with the plasma membrane to release cargo into the cytoplasm. They are broadly effective and widely used in routine applications. FuGENE® (*Alternative B*) and polyeth-



